

Dietary Phosphorus Acutely Impairs Endothelial Function



Excessive dietary phosphorus may increase cardiovascular risk in healthy individuals as well as in patients with chronic kidney disease, but the mechanisms underlying this risk are not completely understood. To determine whether postprandial hyperphosphatemia may promote endothelial dysfunction, we investigated the acute effect of phosphorus loading on endothelial function in vitro and in vivo. Exposing bovine aortic endothelial cells to a phosphorus load increased production of reactive oxygen species, which depended on phosphorus influx via sodium-dependent phosphate transporters, and decreased nitric oxide production via inhibitory phosphorylation of endothelial nitric oxide synthase. Phosphorus loading inhibited endothelium-dependent vasodilation of rat aortic rings. In 11 healthy men, we alternately served meals containing 400 mg or 1200 mg of phosphorus in a double-blind crossover study and measured flow-mediated dilation of the brachial artery before and 2 h after the meals. The high dietary phosphorus load increased serum phosphorus at 2 h and significantly decreased flow-mediated dilation. Flow-mediated dilation correlated inversely with serum phosphorus. Taken together, these findings suggest that endothelial dysfunction mediated by acute postprandial hyperphosphatemia may contribute to the relationship between serum phosphorus level and the risk for cardiovascular morbidity and mortality.